EX. NO: 3: SIMPLE SHELL PROGRAMS

AIM:

To write simple shell scripts using shell programming fundamentals.

DESCRIPTION:

The activities of a shell are not restricted to command interpretation alone. The shell also has Rudimentary programming features. When a group of commands has to be executed regularly, they are stored in a file (with extension .sh). All such files are called shell scripts or shell programs. Shell programs run in interpretive mode.

The original UNIX came with the Bourne shell (sh) and it is universal even today. Then came a plethora of shells offering new features. Two of them, C shell (csh) and Korn shell (ksh) has been well accepted by the UNIX fraternity. Linux offers Bash shell (bash) as a superior alternative to Bourne shell.

Preliminaries

1. Comments in shell script start with #. It can be placed anywhere in a line; the shell ignores contents to its right. Comments are recommended but not mandatory
2. Shell variables are loosely typed i.e. not declared. Their type depends on the value assigned. Variables when used in an expression or output must be prefixed by $.
3. The read statement is shell's internal tool for making scripts interactive.
4. Output is displayed using echo statement. Any text should be within quotes. Escape sequence should be used with –e option.
5. Commands are always enclosed with ` ` (back quotes).
6. Expressions are computed using the expr command. Arithmetic operators are + - a. / %. Meta characters * ( ) should be escaped with a \.
7. Multiple statements can be written in a single line separated by ;
8. The shell scripts are executed using the sh command (sh filename).

Swapping values of two variables

Algorithm

Step 1 : Start
Step 2 : Read the values of a and b
Step 3 : Interchange the values of a and b using another variable t as follows: t = a a = b b = t
Step 4 : Print a and b
Step 5 : Stop

Program (swap.sh) # Swapping values

```
echo -n "Enter value for A : "
read a
echo -n "Enter value for B : "
read b
t=$a
a=$b
b=$t
echo "Values after Swapping"
echo "A Value is $a"
echo "B Value is $b"
```

Output

```
$ sh swap.sh
Enter Value for A:5
Enter Value for B:6
Values after Swapping
A value is 6
B values is 5
$
```

Fahrenheit to Centigrade Conversion

Algorithm

Step 1: Start
Step 2: Read Fahrenheit value
Step 3: Convert Fahrenheit to centigrade using the formulae: (Fahrenheit – 32) × 5/9
Step 4: Print centigrade
Step 5: Stop

Program

```
# Degree conversion
echo -n "Enter Fahrenheit : " read f c=`expr\( $f - 32 \) \* 5 / 9`
echo "Centigrade is : $c"
```

Output

```
$ sh temp.sh
Enter Fahrenheit:4
Centrigrade is: -15
$
```